

YCC workshop pre-read: Current state of play for refrigerant methodologies in Article 5 countries

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Context for this document

This document extends Carbon Containment Lab's 2023 work outlining the landscape of carbon credit methodologies for refrigerants and their upcoming "State of Cooling Credits in the Carbon Market" research by (1) reflecting recent methodological updates, and (2) focusing on methodologies that could be applied in Article 5 (A5) contexts — particularly within Southeast Asia, which has been identified as a high-opportunity area for refrigerant carbon projects.

Background

In 2023, the Carbon Containment Lab [released research](#) on the existing landscape of carbon credit methodologies. Outputs included:

- [Primer on Refrigerant Methodologies](#): Complete summary table of existing fluorocarbon methodologies and eligibility flowchart.
- An associated [one page summary](#) and [flowchart](#).

Through this work, they identified a particular gap within the methodology landscape for HFC destruction outside North America, which prompted them to author a draft methodology for HFC destruction in A5 countries (based in part on Verra's VM0016 methodology).

In the years since, options for carbon crediting in Article 5 countries — where cooling demand continues to rise — have begun to expand. Notably:

- VM0016 has been updated to include HFC destruction under a limited set of circumstances.
- The CCL methodology has been adapted into the [Global Heat Reduction Initiative](#) and is actively being adapted into the [Open Carbon Protocol](#).
- [Isometric](#) has announced plans to issue an HFC destruction methodology.

Both the Open Carbon Protocol and Isometric methodologies are expected to be shared publicly soon and finalized in the coming months.

Cascade Climate has a [particular interest in refrigerant management](#) in Southeast Asia, where they are expanding their programmatic work. That includes a focus on carbon markets as one high-potential lever in the refrigerant transition.

This document presents an updated overview of carbon market methodologies, with particular attention to those relevant to HFC destruction in Article 5 countries and Southeast Asia.

Summary of Methodologies

Active carbon credit methodologies for F-gases in Article 5 countries include:

- [ACR - Destruction of Ozone Depleting Substances from International Sources](#): Covers ODS destruction. Has been used in an Article 5 context (e.g., Tradewater ODS destruction in Thailand).
- [CAR - Article 5 Ozone Depleting Substances Protocol](#): Covers ODS destruction of gases from Article 5 countries, but requires that destruction to occur in the U.S. No credits since 2013, in part due to the transition in focus to ODS destruction projects for use in the California Cap and Trade system.
- **Verra's** [VM0016 - Recovery and Destruction of Ozone-Depleting Substances \(ODS\) from Products](#): Primarily covers ODS destruction, but updated in December 2025 to include HFC destruction, albeit under limited circumstances (e.g., if an Article 5 country has reached 50% of the phasedown schedule, or if there are no reclamation facilities in the country).
- **GHR's** [GHR001 Methodology for Assessing Emissions Reductions via Collection and Destruction of HFCs](#): GHR issues credits for short-lived climate pollutants. Methodology largely follows the proposed CCL approach, with a handful of minor divergences.
- **Thailand's** [T-VER-P-METH-15-01 - Collection of Used Refrigerants for Reclamation and Utilization or Decomposition of Refrigerants](#): Methodology for HFC reclamation or destruction in Thailand, oriented toward international buyers (there is also a domestic equivalent). ICROA-approved.

In addition, there are a handful of methodologies under development or that have been proposed:

- **Oneshot.earth/Open Carbon Protocol** and **Isometric** HFC destruction methodologies — discussed above.
- **JCM** HFC Destruction in Vietnam methodology — [proposed in 2022](#), but it is not clear whether it progressed past the public input phase.

The Open Carbon Protocol and Isometric methodologies are initially designed for the voluntary carbon market (VCM), which mainly serves private sector actors purchasing credits against their sustainability and net zero goals. The JCM methodology is intended for use either against Japan's NDC or under Article 6.2 of the Paris Agreement, which allows for carbon credit trading between countries as a means of potential NDC compliance.

These methodologies cover ODS and HFC destruction and/or reclamation, an essential part of lifecycle refrigerant management where the expected alternative end state is often atmospheric release. Designed correctly, VCM projects within these categories have the potential to help to stand up refrigerant recovery capacity in areas with high refrigerant venting rates — setting the stage for scaled-up support through Article 6 integration and eventual policy and industry uptake. Refrigerant recovery is a key bottleneck in A5 countries where gas venting rates are

extremely high, and carbon market financing for recover-to-destroy projects can support initial recovery capacity build-up in a highly additional way — while avoiding near-term emissions.

To our knowledge, beyond destruction, there are currently no active methodologies that incentivize the transition to advanced refrigerants or finance leak detection via the VCM in the A5 context (with the exception of a natural refrigerant methodology, listed but not publicly available through Thailand's Greenhouse Gas Organization), though methodologies for both exist in North America. Such activities are critically important for reducing greenhouse gas emissions from refrigerants, but are generally seen as in a more nascent stage compared to ODS and HFC destruction projects.

Methodological criteria and quality considerations

All these methodologies aim to ensure that credits are robust, additional, and avoid perverse incentives, but differ significantly in their specifics around which gas sources are allowed, how they mitigate potential project risks, and how credits are generated.

Key credit quality-related challenges include:

- **Avoiding perverse incentives:** Since the [HFC-23 scandal](#) over a decade ago, when factories in China and India boosted their HFC-23 production in order to receive CDM carbon credits, stakeholders have been acutely aware of the need to avoid incentivizing increases in virgin gas production through carbon finance. This remains a concern: at current carbon prices, the income from destroying HFCs is frequently many multiples the price of virgin gas – not factoring in the costs of recovery, destruction/reclamation, or monitoring – so arbitrage might be expected absent effective controls.
- **Additionality:** Carbon credits require assurance that the claimed climate benefits are occurring directly as a result of climate finance. In practice, the counterfactual case is that almost all HFCs are vented to the atmosphere after use in Article 5 countries. However, that could change, particularly as Kigali caps make HFCs more valuable and increase the incentives for recovery, or as countries roll out new management and compliance policies.¹ We also want to avoid a situation in which carbon finance inadvertently disincentivizes robust policy.
- **Leakage and carbon accounting:** Credits need to consider factors like what is serving as the replacement when gas is destroyed (rather than reclaimed); how reclaimed gas interacts with supply quotas; and how supply-demand dynamics might complicate expected benefits (e.g., whether paying for recovered gas might increase increase supply of virgin gas through market dynamics rather than fraud). Data to support these

¹ Even where policy exists, there can be a discrepancy between policies on paper and what is enforced. This is particularly true in the refrigerant space, where it is difficult to enforce venting bans and/or recovery requirements. Carbon projects that support activities required by policies that exist on paper (e.g., venting bans), but are not enforced, can still be additional [according to the ICVCM](#) — provided that the projects are not in high-income countries and that there is “authoritative and up-to-date” information showing that policies are not enforced.

factors can be challenging to come by, particularly in A5 contexts, but is critical to determining credit quality.

Specific methodological criteria that influence how credits are generated, and how reliable they are, include:

- The types of activities allowed (e.g., destruction or reclamation)
- The specific gases allowed (e.g., CFC-11, HFC-32, etc.)
- Where gases can be taken from (e.g., stockpile, servicing, end-of-life)
- When gases and/or projects are allowed to be credited (e.g., when there is no in country production capacity, at a certain portion of Kigali phasedown)
- How additionality is assessed
- Data about the alternative fate of gases (e.g., rates of venting in the country by gas source)
- What assumptions are allowed about alternative gas fates or replacement gases
- Chain of custody requirements (what checks show where the gases come from and what happens to them through the project's lifecycle)
- Price limits and associated documentation
- Checks on the composition of recovered or reclaimed gas
- Verification requirements and cadence
- Timing of credit issuance (upfront upon destruction, or over time compared to the counterfactual baseline)

Looking forward

VCM refrigerant destruction projects — for both ODS and HFCs — have the potential to provide a quick path to successful projects and impact. However, while some methodologies for A5 contexts exist, they have yet to be fully tested across a wide range of contexts and markets. Effective market guardrails necessary for scale — especially for recently published or in-progress HFC destruction methodologies and projects — remain theoretical or under-tested.

The community needs to get these guardrails and pre-conditions right — and test them — if they want to benefit from the recent influx of interest into super pollutant mitigation from the private sector. If successful, these projects can accelerate the LRM transition in A5 countries and build the way for more integrated approaches to meet the needs of a warming world.

Appendix I: Cooling-related methodologies across A5 and non-A5 countries

Methodology	Registry	Type	Category	Active period	Credits ²	A5/SE Asia
Adaptation of Carbon Containment Lab Article 5 HFC destruction methodology	OCP	Voluntary	ODS and HFC destruction	<i>In progress</i>	0	Yes
ACR Destruction of Ozone Depleting Substances from International Sources	ACR	Voluntary	ODS	2021 - Present	2,140,672	Yes
VM0016 Recovery and Destruction of Ozone-Depleting Substances (ODS) from Products	VCS	Voluntary	ODS and HFC destruction	2017 - Present	342,244	Yes
CAR Article 5 Ozone Depleting Substances Protocol	CAR	Voluntary	ODS	2012 - Present	4,009,926	Yes ³
Introduction of HFCs destruction facilities in Viet Nam	JCM	Article 6	HFC destruction	To come	0	Vietnam
T-VER-S-METH-15-03 Use of Reclaimed Refrigerant [domestic use]	T-VER	Voluntary	HFC reclamation	2025 - Present	None known	Thailand
T-VER-P-METH-15-01 [Premium - for international use, ICROA-approved]	T-VER	Voluntary	HFC reclamation or dest.	2025 - Present	None known	Thailand
ACR Destruction of Ozone Depleting Substances and High-GWP Foam	ACR	Voluntary	ODS	2017 - Present	1,024,934	Not currently
ACR Certified Reclaimed HFC Refrigerants, Propellants, and Fire Suppressants	ACR	Voluntary	HFC reclamation	2015 - Present	25,533,075	Not currently
ACR Advanced Refrigeration Systems	ACR	Voluntary	HFC transition	2015 - Present	4,393,572	Not currently
CAR Mexico Halocarbon Protocol	CAR	Voluntary	ODS and HFC destruction	2021 - Present	0	Not currently
CAR U.S. Ozone Depleting Substances Protocol	CAR	Voluntary	ODS	2012 - Present	65,808	Not currently
VM0001 Refrigerant Leak Detection	VCS	Voluntary	HFC leak avoidance	2012 - Present	259,102	Not currently
ARB Compliance Offset Protocol: Destruction of U.S. Ozone Depleting Substances Banks	ACR, CAR	Compliance - CA	ODS	2011 - Present	27,179,229	Not currently
ECY Compliance Offset Protocol: Ozone Depleting Substances Projects (ODS)	ACR (ECY)	Compliance - WA	ODS	2025 - Present	590,281	Not currently
Reducing greenhouse gas emissions from refrigeration systems (1.2)	Canada	Compliance - Canada	ODS	2023 - Present	7697	Not currently
AM0001 Decomposition of fluoroform (HFC-23) waste streams	VCS	Voluntary	HFC destruction	<i>Inactivated 2024</i>	4,483,537	No
ACR Transition to Advanced Formulation Blowing Agents in Foam Manufacturing and Use	ACR	Voluntary	HFC transition	<i>Inactivated 2024</i>	45,479,139	No

² Via Berkeley Carbon Trading Project Voluntary Registry Offsets Database - v2025-08. Exact data may vary slightly by source.

³ Gases can come from anywhere, but must be destroyed in the United States.